Calibo AI Academy · Path 2 · KLU Batch

MUC02 — Trainer Briefing

Credit Pre-Screening Engine — Loan Intake Optimisation

Phase 2 · Supervised Learning Part 1 · BFSI — Retail Lending

15–30 min	7 steps	2 files	BFSI
Briefing length	Analytical tasks	To share	Industry

Who you are building this model for

A mid-size private bank receives about 3,000 personal loan applications a month. Every one gets a 25-minute manual review, and nearly half end up declined — many for reasons visible in the first minute. The Head of Retail Lending has asked the Credit Policy Analyst: can a model pre-screen applications so analysts spend their time where judgement is needed?

She needs two things from you:

1. A model the credit team can trust

What drives decisions today? How well can a model predict them? Where does it fail?

2. A threshold the bank can defend

How many applications can be auto-declined while wrongly declining no more than 5% of good customers — and what is that worth in analyst hours?

"The day you understand the value of the work you do,
is the day you add more value to the work you do."

What this means for this use case:

1. A confusion matrix is not a score. Every false positive is a creditworthy customer turned away. Students should be able to say how many.
2. The threshold is a business decision. The bank sets the risk limit; the model only tells you where that limit falls.
3. Fairness is part of the job, not an add-on. Leaving gender out of the model does not, on its own, make the model fair.

SEVEN STEPS

One continuous story — each step builds on the previous

5.1 Load, inspect & clean

Missing incomes, blank credit scores, 12 annual incomes. Every decision written down.

5.2 What drives approval?

Approval by credit score band and employment type. Find the 650 cliff.

5.3 Affordability features ★

Build FOIR, loan-to-income, NTC flag. Find the FOIR cliff.

5.4 Logistic regression

Predict Declined. Confusion matrix and coefficients in business language.

5.5 Auto-decline threshold ★

Threshold table, 5% wrong-decline rule, applications and hours per month.

5.6 Fairness check

Approval by gender, within employment type, and from the model.

5.7 Credit Policy Briefing ★

4–6 sentences. Threshold, hours, drivers, NTC caveat, fairness.

★ *Most important steps*

Before any code — 15 to 20 minutes

Complete the canvas before touching the data. This is not optional.

Situation (S)

~3,000 applications a month, each reviewed manually for ~25 minutes.

Complication (C)

Analyst time spent on clear-cut declines; good applicants wait longer.

Question (Q)

Not "build a model" — which applications can safely leave manual review, and at what threshold?

SSOT

The dataset — and the decision column is past analyst judgement, not repayment.

Success definition

A threshold within the 5% risk limit and a stated number of hours freed.

Stakeholder

Credit Policy Analyst — and the analysts and applicants affected.

The difference between passing and distinction

Area	What we often see	What we want to see
Cleaning	Rows with blank credit scores dropped	NTC applicants kept and flagged — 9% of all applicants
Model	"The model is 79% accurate"	"At 0.5, 225 good applicants would be wrongly declined"
Threshold	Left at 0.5	"0.90 — 15% auto-declined, 4.7% wrong, ~190 analyst hours a month"
Fairness	"We did not use gender, so it is fair"	"The gap comes from income and job mix — monitor outcomes by gender after launch"

Two files — both go in the same folder as the notebook

Student document

MUC02_KLU_Credit_PreScreening.docx — full instructions, deliverables, CBIM canvas and submission checklist

Dataset

MUC02_Credit_PreScreening_Dataset.csv — 12,000 applications · 19 columns · Apr–Sep 2026 · cities across India

Setup — if not already installed:

pip install pandas numpy matplotlib seaborn scikit-learn jupyter

What to submit

Individual

- [YourName]_MUC02_KLU_Notebook.ipynb
- CBIM canvas at top → Steps 5.1 through 5.7
- Runs end-to-end — kernel restart + run all cells tested
- [YourName]_MUC02_KLU_Summary.docx (max 400 words)

Group of 2–3

- [TeamName]_MUC02_KLU_Notebook.ipynb — each step credits who built it
- CBIM canvas completed together — all members contributed
- [TeamName]_MUC02_KLU_Summary.docx (max 500 words)
- 150-word individual reflection from each member — submitted separately

Deadline: to be confirmed — update the student document before distributing.

Three things before you start

1

CBIM canvas first. Every time. Before any code.

2

The threshold is the answer. Use the table, apply the 5% rule, and state what it frees up in applications and hours.

3

Some applicants should always see a human. Find out who, and say so in your briefing.

Any questions before we begin?